

Estimation of ribosome concentration in *Prochlorococcus* cells

We used *Prochlorococcus* cultures with a cell concentration of 3×10^7 cells/ml. Our cell extracts had a protein concentration of 3.2 mg protein/ml cell extract. Since we obtained a volume of 0.5 mL of cell extract from 1 L of *Prochlorococcus* cultures, we had a protein concentration of 1.6 mg of total protein in 1 L of culture, or 0.0016 mg per ml of culture.

Thus we have 0.0016 mg of protein in 30.000.000 cells; so 1 mg of total protein corresponds to 1.875×10^{10} cells.

The average concentration of ribosomal proteins in our control samples is 122.16 pmoles/mg of protein. Assuming there is 1 ribosomal protein per ribosome, this means 122.16×10^{-12} moles ribosomes $\times 6.023 \times 10^{23}$ ribosomes/mol = 7.36×10^{13} ribosomes in 1 mg of total protein. Hence, we have 7.36×10^{23} ribosomes in 1.875×10^{10} cells, which corresponds to **3,924 ribosomes per cell in control samples**.

The average concentration of ribosomal proteins in our azaserine-treated samples is 52.34 pmoles/mg protein. By doing the same calculation described above, we have 52.34×10^{-12} moles ribosomes $\times 6.023 \times 10^{23}$ ribosomes/mol = 3.15×10^{13} ribosomes in 1 mg of total protein. Hence we have 3.15×10^{13} ribosomes in 1.875×10^{10} cells, which corresponds to **1,681 ribosomes per cell in azaserine-treated samples**.